- **Small finite magmas**, which can be described explicitly by multiplication tables, could be tested by brute force computations to provide a large number of finite counterexamples to implications, or by ATP-assisted methods. See Section 5.1.
- **Linear models**, in which the magma operation took the form $x \diamond y = ax + by$ for some (commuting or noncommuting) coefficients a, b, allowed for another large class of counterexamples to implications, which could be automatically scanned for, either by brute force or by Gröbner basis type calculations; many of these examples could also be made finite. See Section 5.2.
- **Translation invariant models**, in which the magma operation took the form $x \diamond y = x + f(y - x)$ on an additive group, or $x \diamond y = xf(x^{-1}y)$ on a noncommutative group, reduce matters to analyzing certain functional equations; see Section 5.3.
- To each equation E one can associate a "**twisting semigroup**" S_{E}. If S_{E} is larger than $S_{\mathrm{E}'}$, then this can often be used to disprove the implication $\mathrm{E} \models \mathrm{E}'$; see Section 5.4.
- **Greedy methods**, in which either the multiplication table $(x, y) \mapsto x \diamond y$ or the function f determining a translation-invariant model are iteratively constructed by a greedy algorithm subject to a well-chosen ruleset, were effective in resolving many implications not easily disposed of by preceding methods. See Section 5.5.
- Starting with a simple base magma $\mathcal{M}$ satisfying both E and E′, and either **enlarging** it to a larger magma $\mathcal{M}'$ containing $\mathcal{M}$ as a submagma, **extending** it to a magma $\mathcal{N}$ with a projection homomorphism $\pi : \mathcal{N} \to \mathcal{M}$, or *modifying the multiplication table* on a small number of values, also proved effective when combined with greedy methods or with a "**magma cohomology**" construction. See Section 5.6.
- **Syntactic methods**, such as observing a "matching invariant" of the law E that was not shared by the law E′, could be used to obtain some refutations. For instance, if both sides of E had the same order, but both sides of E′ did not, this could be used to syntactically refute $\mathrm{E} \models \mathrm{E}'$. Similarly, if the law E was confluent, enjoyed a complete rewriting system, or otherwise permitted some understanding of the free magma associated to that law, one could decide the assertions $\mathrm{E} \models \mathrm{E}'$ for all possible laws E′, or at least a significant fraction of such laws. We discuss these methods, and the extent to which they can be automated, in Section 6.
- Some **ad hoc models** based on existing mathematical objects, such as infinite trees, rings of polynomials, or "Kisielewicz models" utilizing the prime factorization of the natural numbers, could also handle some otherwise difficult cases. In some cases, the magma law induced some relevant and familiar structures, such as a directed graph or a partial order, which also helped guide counterexample constructions. We will not detail these diverse examples here, but refer the reader to the ETP blueprint for more discussion.
- **Automated theorem provers** were helpful in identifying which simplifying axioms could be added to the magma without jeopardizing the ability to refute the desired implication $\mathrm{E} \models \mathrm{E}'$ or $\mathrm{E} \models_{\mathrm{fin}} \mathrm{E}'$.

While the vast majority of negative implications could be quickly resolved by one of the above techniques, either with human input or in a completely automated fashion, there were perhaps two dozen such negative results that required quite delicate and *sui generis* constructions. The hardest such implication, $\mathrm{E1729} \not\models \mathrm{E817}$, took several months to establish

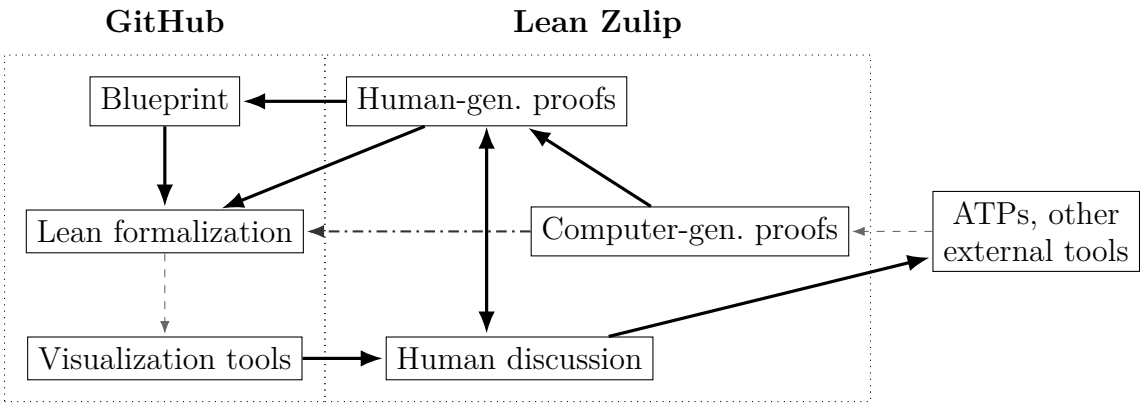

FIGURE 4. Some of the main dynamics in which proofs were generated, discussed within the Lean Zulip channel and then formalized in the GitHub repository. Boldface arrows indicate human activities, such as proposing an automated attack on outstanding implications, converting a computer-generated proof into a human-readable format, formalizing a human readable proof directly, or first creating a more precise blueprint for other collaborators to work on. Dashed arrows indicate fully automated processes, while the partly dashed line indicates a semi-automated process requiring human supervision.

and then formalize (using a combination of many of the above constructions), with the final proof in *Lean* requiring just over 4000 dedicated lines of code from multiple contributors.

In the course of completing the implication graph, some interesting new algebraic structures were discovered. One such example concerns the magmas satisfying E1485, which we refer to as *weak central groupoids* as they contain the central groupoids (satisfying E168) as a subclass. In [35] it was observed that all finite central groupoids have order equal to a perfect square n^2; empirically, we have found that finite weak central groupoids always have order n^2 or $2n^2$, although we have no rigorous proof of this claim; they also have a graph-theoretic interpretation analogous to the interpretation of central groupoids as digraphs with the unique path property. For these and other observations we refer the reader to the blueprint of the ETP.

The objective of using the data from the ETP to establish well-calibrated benchmarks to evaluate ATPs remains an interesting open problem; the participants of this project did not have the required expertise to develop and test such benchmarks to the standards expected in the area. However, in Section 7 we present a more informal "field report" of our experiences using ATPs in the project, in the hope that this will provide some useful guidance to other researchers seeking to incorporate ATPs into their own research.

1.4. **Further directions.** While the primary objective of the ETP was being completed, some additional related results were generated as spinoffs. Specifically:

- In the blueprint on the ETP web site, we report some partial progress on classifying which of the 57 882 distinct laws of order 5 are equivalent to the singleton law E2, either with or without the requirement that the magma be finite.

- In Section 9 we report on the determination of laws with full spectrum, i.e., with magmas satisfying them of all finite sizes.
- In Section 10 we report on classifying the laws of order 8 that are equivalent to the Higman-Neumann law E42323216.

We refer the reader to the ETP blueprint for a more in depth discussion of several of the topics covered in this report, including a detailed treatment of particularly difficult equations and complex magma constructions, further treatment of term rewriting theory and equational law invariants, a list of "hard" implications that required particular effort to resolve in our project, and some implementation details of how tools such as automated theorem provers were utilized.

2. Notation and mathematical foundations

If $\mathcal{M} = (M, \diamond)$ is a magma, we define the left and right multiplication operators $L_a, R_a \colon M \to M$ for $a \in M$ by the formula

$$L_x y = R_y x \coloneqq x \diamond y. \tag{1}$$

We also define the squaring operator $S \colon M \to M$ by

$$Sx \coloneqq x \diamond x = L_x x = R_x x. \tag{2}$$

A *homomorphism* $f \colon \mathcal{M} \to \mathcal{M}'$ between two magmas $\mathcal{M} = (M, \diamond)$, $\mathcal{M}' = (M', \diamond')$ is a function $f \colon M \to M'$ such that $f(x \diamond y) = f(x) \diamond' f(y)$ for all $x, y \in M$. An *isomorphism* is a homomorphism that is invertible (which implies that the inverse is also a homomorphism). An *endomorphism* is a homomorphism from a magma to itself.

If X is an alphabet, we let $\mathcal{M}_X = (M_X, \diamond)$ denote the free magma generated by X, thus an element of $\mathcal{M}_X$ is either a letter in X, or of the form[5] $w_1 \diamond w_2$ with $w_1, w_2 \in M_X$. Every function $f \colon X \to M$ into a magma $\mathcal{M} = (M, \diamond)$ extends to a unique homomorphism $\varphi_f \colon \mathcal{M}_X \to \mathcal{M}$. Formally, an equational law with some indeterminates in X can be written as $w_1 \simeq w_2$ for some $w_1, w_2 \in M_X$; a magma $\mathcal{M} = (M, \diamond)$ then satisfies this law if and only if $\varphi_f(w_1) = \varphi_f(w_2)$ for all $f \colon X \to M$. We also define the order of a word $w \in M_X$ to be the number of occurrences of $\diamond$ in the word, thus letters in X are of order 0, and the order of $w_1 \diamond w_2$ is the sum of the orders of w_1, w_2, plus one.

A *theory* is a collection Γ of equational laws; we say that a magma $\mathcal{M}$ *satisfies* a theory, and write $\mathcal{M} \models \Gamma$, if every law in Γ is satisfied by $\mathcal{M}$. If E is an equational law, we write $\Gamma \models \mathrm{E}$ if every magma that satisfies Γ also satisfies E. A *free magma* $\mathcal{M}_{X,\Gamma} = (M_{X,\Gamma}, \diamond)$ for such a theory Γ and an alphabet X is a magma satisfying Γ together with a map $\iota_{X,\Gamma} \colon X \to M_{X,\Gamma}$ which is universal in the sense that every function $f \colon X \to \mathcal{M}$ to a magma $\mathcal{M}$ satisfying Γ uniquely determines a homomorphism $\varphi_{f,\Gamma} \colon \mathcal{M}_{X,\Gamma} \to \mathcal{M}$ such that $\varphi_{f,\Gamma} \circ \iota_{X,\Gamma} = f$. This magma is unique up to isomorphism; a canonical way to construct it is as the quotient $\mathcal{M}_X / \sim_\Gamma$ of the free magma $\mathcal{M}_X$ by the equivalence relation $\sim_\Gamma$ given by declaring $w \sim_\Gamma w'$

[5] Strictly speaking, one should use parentheses and write $(w_1 \diamond w_2)$ to avoid ambiguity, but to reduce clutter we shall abuse notation by omitting parentheses when no ambiguity is caused by doing so.